RAFFLES INSTITUTION
RAFFLES PROGRAMME 2026
YEAR 3 MATHEMATICS 2
TEST 2 MOCK PRACTICE PAPER 4

20

Parent's Signature | Duration: 30 minutes

Name: ________________________________ ( ) Class: 3 ( ) Date: ___________

Instructions:

1. Answer **all** questions in the space provided.
2. This paper consists of printed pages.
3. The use of a calculator is allowed unless otherwise stated.
4. Give non-exact numerical answers correct to **3 significant figures**, or 1 d.p. for angles, unless stated otherwise.
5. Omission of essential working will result in loss of marks.

1. Solve the equation

$$(\lg x)^2 + \lg(x^3) = 10,$$

giving your answers in exact form.

[4]

Working space

2. Solve the equation

$$5^x \cdot 3^{2x+1} = 7^{x-1},$$

giving your answer correct to 3 significant figures.

[3]

Working space

3.

(i) Sketch the graph of $y = \ln(2x - 1)$, indicating clearly all axial intercepts and the asymptote. **[2]**

Sketch here

(ii) By drawing a suitable straight line on your sketch in **(i)**, find the number of solutions to the equation **[3]**

$$\sqrt{(2x-1)} / e^{2x} = e^{-3}.$$

Working space

4. Solve the simultaneous equations

$$2 \log_3 x - \log_3 y = 1$$

$$\log_3 x + 2 \log_3 y = 8.$$

[5]

Working space

~ End of Paper ~

WORKED SOLUTIONS — MOCK PAPER 4

Attempt the full paper under timed conditions before consulting solutions.

Question 1 [4 marks]

M1 Power law: $\lg(x^3)=3\lg x$. Let $u=\lg x$: $u^2+3u-10=0$.

M1 $(u+5)(u-2)=0 \Rightarrow u=-5$ or $u=2$.

A1 $\lg x=-5 \Rightarrow \boldsymbol{x=10^{-5}}$.

A1 $\lg x=2 \Rightarrow \boldsymbol{x=100}$.

Why this works: The power law converts $\lg(x^3)$ to $3\lg x$, giving a true quadratic in $\lg x$. Both roots produce valid values of x (both positive), so neither is rejected.
Common error: Writing $(\lg x)^2=\lg(x^2)=2\lg x$. This is WRONG. $(\lg x)^2$ means squaring the entire log value; the power law only applies inside the log argument.

Question 2 [3 marks]

M1 Take lg: $x\lg5+(2x+1)\lg3=(x-1)\lg7$. Expand: $x(\lg5+2\lg3-\lg7)=-\lg7-\lg3$.

M1 $x(\lg 45/7)=-\lg21$. Compute: $\lg(45/7)\approx0.808$, $\lg21\approx1.322$.

A1 $x=-1.322/0.808=\mathbf{-1.64}$ (3 s.f.).

Why this works: With x in multiple exponents with different bases, taking lg of both sides and expanding is the only route. The five moves are always: take lg, expand, collect x terms, factor, divide.

Question 3 [5 marks]

(i) Asymptote: $2x-1=0 \Rightarrow x=1/2$. x-intercept: $2x-1=1 \Rightarrow x=1$. No y-intercept (domain $x>1/2$). Increasing curve.

M1 **(ii)** Take ln: $(1/2)\ln(2x-1)-2x=-3$. Multiply by 2: $\ln(2x-1)=4x-6$.

M1A1 Draw line $\boldsymbol{y=4x-6}$. Slope 4, x-intercept 3/2. The line intersects $y=\ln(2x-1)$ at **2 points**.

Question 4 [5 marks]

M1 Let $X=\log_3 x$, $Y=\log_3 y$. System: $2X-Y=1$ and $X+2Y=8$.

M1 From first: $Y=2X-1$. Sub: $X+2(2X-1)=8 \Rightarrow 5X=10 \Rightarrow X=2$, $Y=3$.

A1 $\log_3 x=2 \Rightarrow \boldsymbol{x=9}$.

A1 $\log_3 y=3 \Rightarrow \boldsymbol{y=27}$.

A1 Verify: $2(2)-3=1$ ✔ and $2+2(3)=8$ ✔.